

WIPRO TALENTNEXT PROGRAM

PBL HANDS-ON ASSIGNMENT

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Topic: Restricting and Sorting Data

Question 1

Display the Ename, Sal, comm from EMP table who earn a commission, sort the records in descending order of salary and comm. Use column's numeric position in the ORDER BY clause.

Query:

```
CREATE TABLE EMP
```

```
(
```

```
    EMPNO NUMBER(4),
```

```
    ENAME VARCHAR2(10),
```

```
    JOB VARCHAR2(10),
```

```
    MGR NUMBER(4),
```

```
    HIREDATE DATE,
```

```
    SAL NUMBER(7,2),
```

```
    COMM NUMBER(7,2),
```

```
    DEPTNO NUMBER(2)
```

```
);
```

```
INSERT INTO EMP VALUES (7000,'DEEPU','CLERK',7902,DATE '1980-12-17',800,NULL,20);
```

```
INSERT INTO EMP VALUES (7001,'ASHU','SALESMAN',7698,DATE '1981-02-20',1600,300,30);
```

```
INSERT INTO EMP VALUES (7002,'BHANU','SALESMAN',7698,DATE '1981-02-22',1250,500,30);
```

```
INSERT INTO EMP VALUES (7003,'JONES','MANAGER',7839,DATE '1981-04-02',2975,NULL,20);
```

```
INSERT INTO EMP VALUES (7004,'MANI','SALESMAN',7698,DATE '1981-09-28',1250,1400,30);
```

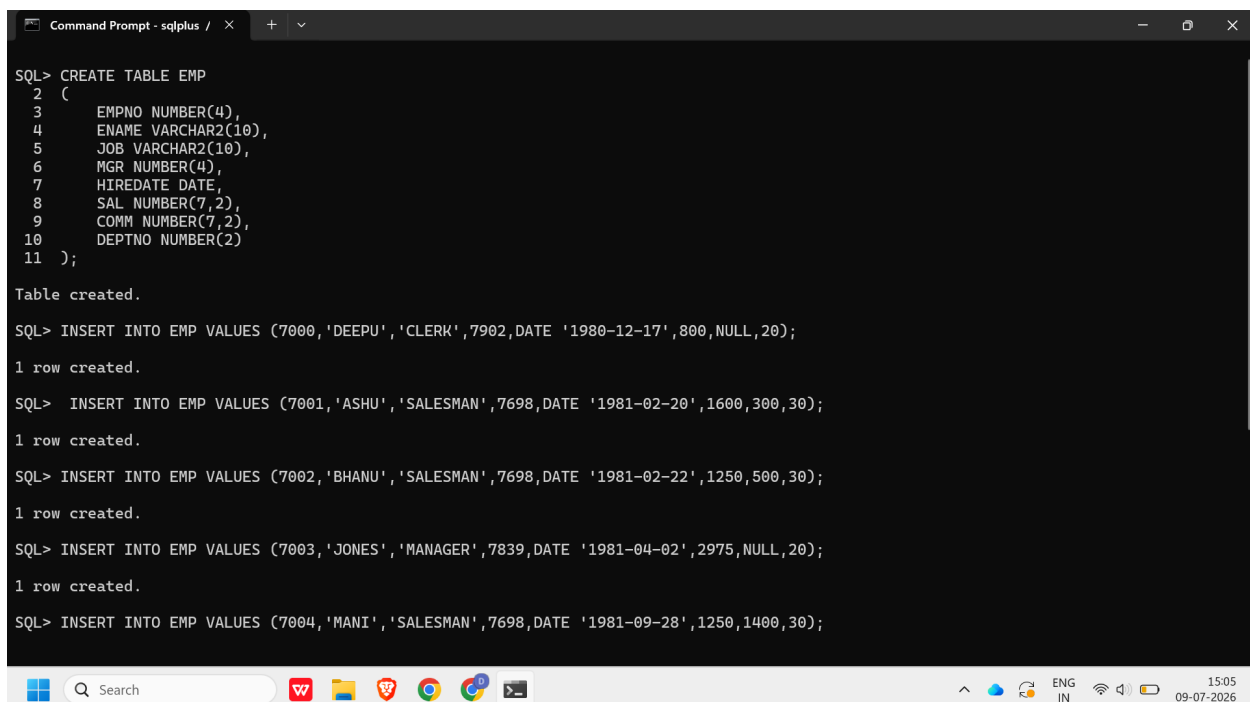
```
INSERT INTO EMP VALUES (7698,'BALU','MANAGER',7839,DATE '1981-05-01',2850,NULL,30);
```

```
INSERT INTO EMP VALUES (7788,'SCOTT','ANALYST',7566,DATE '1987-04-19',3000,NULL,20);
```

```
INSERT INTO EMP VALUES (7844,'TURNER','SALESMAN',7698,DATE '1981-09-08',1500,0,30);
```

```
INSERT INTO EMP VALUES (7900,'JAMES','CLERK',7698,DATE '1981-12-03',950,NULL,30);
```

```
SELECT ENAME, SAL, COMM FROM EMP WHERE COMM IS NOT NULL ORDER BY 2 DESC, 3 DESC;
```



```
SQL> CREATE TABLE EMP
2 (
3     EMPNO NUMBER(4),
4     ENAME VARCHAR2(10),
5     JOB VARCHAR2(10),
6     MGR NUMBER(4),
7     HIREDATE DATE,
8     SAL NUMBER(7,2),
9     COMM NUMBER(7,2),
10    DEPTNO NUMBER(2)
11 );

Table created.

SQL> INSERT INTO EMP VALUES (7000,'DEEPU','CLERK',7902,DATE '1980-12-17',800,NULL,20);

1 row created.

SQL> INSERT INTO EMP VALUES (7001,'ASHU','SALESMAN',7698,DATE '1981-02-20',1600,300,30);

1 row created.

SQL> INSERT INTO EMP VALUES (7002,'BHANU','SALESMAN',7698,DATE '1981-02-22',1250,500,30);

1 row created.

SQL> INSERT INTO EMP VALUES (7003,'JONES','MANAGER',7839,DATE '1981-04-02',2975,NULL,20);

1 row created.

SQL> INSERT INTO EMP VALUES (7004,'MANI','SALESMAN',7698,DATE '1981-09-28',1250,1400,30);
```

```
Command Prompt - sqlplus / X + v
SQL> SELECT ENAME, SAL, COMM FROM EMP WHERE COMM IS NOT NULL ORDER BY 2 DESC, 3 DESC;

ENAME          SAL      COMM
-----
ASHU            1600       300
TURNER          1500        0
MANI            1250      1400
BHANU           1250       500

SQL>
```

Question 2

The HR department needs a query to display all unique job codes from the EMP table.

Query:

```
SELECT DISTINCT JOB FROM EMP;
```

```
SQL> SELECT DISTINCT JOB FROM EMP;

JOB
-----
CLERK
SALESMAN
ANALYST
MANAGER

SQL> |
```

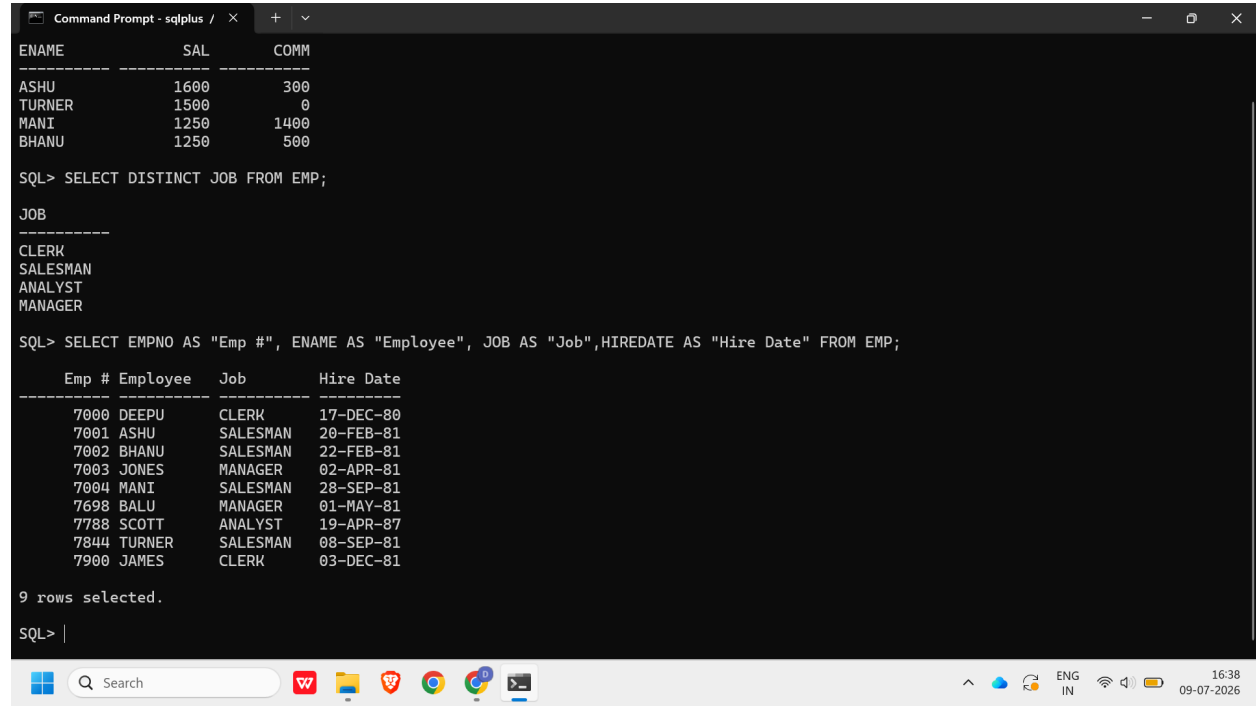
Question 3

The HR department wants more descriptive column headings for its report on employees. Name the column headings Emp #, Employee, Job, and Hire Date, respectively by giving Column Alias.

Query:

```
SELECT EMPNO AS "Emp #", ENAME AS "Employee", JOB AS "Job", HIREDATE AS "Hire Date" FROM EMP;
```

Output:



The screenshot shows a Windows Command Prompt window titled "Command Prompt - sqlplus /". The window displays the output of an SQL query. The first query, `SELECT DISTINCT JOB FROM EMP;`, returns a list of job titles: CLERK, SALESMAN, ANALYST, and MANAGER. The second query, `SELECT EMPNO AS "Emp #", ENAME AS "Employee", JOB AS "Job", HIREDATE AS "Hire Date" FROM EMP;`, returns a table with 9 rows. The table has four columns: Emp #, Employee, Job, and Hire Date. The data is as follows:

Emp #	Employee	Job	Hire Date
7000	DEEPU	CLERK	17-DEC-80
7001	ASHU	SALESMAN	20-FEB-81
7002	BHANU	SALESMAN	22-FEB-81
7003	JONES	MANAGER	02-APR-81
7004	MANI	SALESMAN	28-SEP-81
7698	BALU	MANAGER	01-MAY-81
7788	SCOTT	ANALYST	19-APR-87
7844	TURNER	SALESMAN	08-SEP-81
7900	JAMES	CLERK	03-DEC-81

Below the table, it says "9 rows selected." and the prompt "SQL> |" is visible. The Windows taskbar at the bottom shows the search bar, several application icons, and system tray icons including the date and time (16:38, 09-07-2026).

Question 4

The HR department has requested a report of all employees and their job IDs. Display the last name concatenated with the job ID (separated by a comma and space) and name the column Employee and Title by giving Column Alias.

Query:

```
SELECT ENAME || ', ' || JOB AS "Employee and Title" FROM EMP;
```

Output:

```
SQL> SELECT ENAME || ', ' || JOB AS "Employee and Title"
2 FROM EMP;
```

Employee and Title

```
-----
DEEPU, CLERK
ASHU, SALESMAN
BHANU, SALESMAN
JONES, MANAGER
MANI, SALESMAN
BALU, MANAGER
SCOTT, ANALYST
TURNER, SALESMAN
JAMES, CLERK
```

9 rows selected.

SQL>

Question 5

To familiarize yourself with the data in the EMP table, create a query to display all the data from that table. Separate each column output by a comma. ENAME, JOB, HIREDATE, MGR. Name the column title THE_OUTPUT.

Query:

```
SELECT ENAME || ',' || JOB || ',' || HIREDATE || ',' || MGR AS "THE_OUTPUT" FROM
EMP;
```

Output:

```
SQL> SELECT ENAME || ', ' || JOB || ', ' || HIREDATE || ', ' || MGR AS "THE_OUTPUT" FROM EMP;
```

THE_OUTPUT

```
-----
DEEPU, CLERK, 17-DEC-80, 7902
ASHU, SALESMAN, 20-FEB-81, 7698
BHANU, SALESMAN, 22-FEB-81, 7698
JONES, MANAGER, 02-APR-81, 7839
MANI, SALESMAN, 28-SEP-81, 7698
BALU, MANAGER, 01-MAY-81, 7839
SCOTT, ANALYST, 19-APR-87, 7566
TURNER, SALESMAN, 08-SEP-81, 7698
JAMES, CLERK, 03-DEC-81, 7698
```

9 rows selected.

SQL> |

Question 6

Create a report to display the ename, job, and Hiredate for the employees name is SCOTT or TURNER. Order the query in ascending order by hiredate.

Query:

```
SELECT ENAME, JOB, HIREDATE FROM EMP WHERE ENAME = 'SCOTT' OR ENAME = 'TURNER' ORDER BY HIREDATE;
```

OR

```
SELECT ENAME, JOB, HIREDATE FROM EMP WHERE ENAME IN ('SCOTT' OR 'TURNER') ORDER BY HIREDATE;
```

```
SQL> SELECT ENAME,JOB,HIREDATE FROM EMP WHERE ENAME = 'SCOTT' OR ENAME = 'TURNER' ORDER BY HIREDATE;
```

ENAME	JOB	HIREDATE
TURNER	SALESMAN	08-SEP-81
SCOTT	ANALYST	19-APR-87

```
SQL> SELECT ENAME ,JOB,HIREDATE FROM EMP WHERE ENAME IN ('SCOTT','TURNER') ORDER BY HIREDATE;
```

ENAME	JOB	HIREDATE
TURNER	SALESMAN	08-SEP-81
SCOTT	ANALYST	19-APR-87

Question 7

Display the ename and department number of all employees in departments 20 or 30 in ascending alphabetical order by ename

Query:

```
SELECT ENAME,DEPTNO FROM EMP WHERE DEPTNO IN (20,30) ORDER BY ENAME;
```

Output:

```
SQL> SELECT ENAME,DEPTNO FROM EMP WHERE DEPTNO IN (20,30) ORDER BY ENAME;

ENAME          DEPTNO
-----
ASHU            30
BALU            30
BHANU           30
DEEPU           20
JAMES           30
JONES           20
MANI            30
SCOTT           20
TURNER          30

9 rows selected.

SQL>
```

Question 8

Modify the previous query to display the last name and salary of employees who earn between 2000 and 3000 and are in department 20 or 30. Label the columns Employee and Monthly Salary, respectively giving Column Alias.

Query:

```
SELECT ENAME AS "Employee",SAL AS "Monthly Salary" FROM EMP WHERE SAL
BETWEEN 2000 AND 3000 AND DEPTNO IN (20,30);
```

Output:

```
SQL> SELECT ENAME AS "Employee",SAL AS "Monthly Salary" FROM EMP WHERE SAL BETWEEN 2000 AND 3000 AND DEPTNO IN (20,30);

Employee      Monthly Salary
-----
JONES          2975
BALU           2850
SCOTT          3000

SQL>
```

Question 9

The HR department needs a report that displays the last name and hire date for all employees who were hired in 1981.

Query:

```
SELECT ENAME, HIREDATE FROM EMP WHERE HIREDATE BETWEEN DATE '1981-
01-01' AND DATE '1981-12-31';
```

Output:

```
SQL> SELECT ENAME, HIREDATE FROM EMP WHERE HIREDATE BETWEEN DATE '1981-01-01' AND DATE '1981-12-31';
```

ENAME	HIREDATE
ASHU	20-FEB-81
BHANU	22-FEB-81
JONES	02-APR-81
MANI	28-SEP-81
BALU	01-MAY-81
TURNER	08-SEP-81
JAMES	03-DEC-81

7 rows selected.

```
SQL> SELECT ENAME, HIREDATE FROM EMP WHERE EXTRACT(YEAR FROM HIREDATE) = 1981;
```

ENAME	HIREDATE
ASHU	20-FEB-81
BHANU	22-FEB-81
JONES	02-APR-81
MANI	28-SEP-81
BALU	01-MAY-81
TURNER	08-SEP-81
JAMES	03-DEC-81

7 rows selected.

Question 10

Display the Ename, Sal of employees who earn more than an amount the user specifies after a Prompt.

Query:

```
SELECT ENAME, SAL FROM EMP WHERE SAL > &AMOUNT;
```

Output:

```
SQL> SELECT ENAME, SAL FROM EMP WHERE SAL > &AMOUNT;
```

```
Enter value for amount: 2500
```

```
old 1: SELECT ENAME, SAL FROM EMP WHERE SAL > &AMOUNT
```

```
new 1: SELECT ENAME, SAL FROM EMP WHERE SAL > 2500
```

ENAME	SAL
JONES	2975
BALU	2850
SCOTT	3000

Question 11

Create a report to display the last name and job title of all employees who do not have a manager.

Query:

```
SELECT ENAME,JOB FROM EMP WHERE MGR IS NULL;
```

Output:

```
SQL> SELECT ENAME,JOB FROM EMP WHERE MGR IS NULL;

ENAME      JOB
-----
AISHU      MANAGER
```

Question 12

Create a query that prompts the user for Manager ID and generate EMPNO, ENAME, SAL, DEPTNO. The user should have the ability to sort the records on a selected Column.

```
SELECT EMPNO,ENAME,SAL,DEPTNO FROM EMP WHERE MGR=&MGR ORDER
BY&COLUMN;
```

Output:

```
SQL> SELECT EMPNO,ENAME,SAL,DEPTNO FROM EMP WHERE MGR=&MGR ORDER BY &COLUMN;
Enter value for mgr: 7698
Enter value for column: ENAME
old 1: SELECT EMPNO,ENAME,SAL,DEPTNO FROM EMP WHERE MGR=&MGR ORDER BY &COLUMN
new 1: SELECT EMPNO,ENAME,SAL,DEPTNO FROM EMP WHERE MGR=7698 ORDER BY ENAME

EMPNO ENAME      SAL  DEPTNO
-----
7001  ASHU          1600    30
7002  BHANU         1250    30
7900  JAMES          950     30
7004  MANI          1250    30
7844  TURNER        1500    30
```

Question 13

The HR department wants to run reports based on a manager. Create a query that prompts the user for a MGR and generates the empno, ename, salary, and department for that manager's employees. The HR department wants the ability to sort the report on a selected column.

Query:

```
SELECT EMPNO,ENAME,SAL,DEPTNO FROM EMP WHERE MGR=&MGR ORDER BY  
&COLUMN;
```

Output:

```
SQL> SELECT EMPNO,ENAME,SAL,DEPTNO FROM EMP WHERE MGR=&MGR ORDER BY &COLUMN;  
Enter value for mgr: 7839  
Enter value for column: SAL  
old 1: SELECT EMPNO,ENAME,SAL,DEPTNO FROM EMP WHERE MGR=&MGR ORDER BY &COLUMN  
new 1: SELECT EMPNO,ENAME,SAL,DEPTNO FROM EMP WHERE MGR=7839 ORDER BY SAL  
  
      EMPNO  ENAME      SAL      DEPTNO  
-----  
      7698  BALU      2850      30  
      7003  JONES      2975      20  
  
SQL> |
```

Question 14

Display all employee last names in which the third letter of the name is A.

Query:

```
SELECT ENAME FROM EMP WHERE ENAME LIKE '__A%';
```

Output:

```
SQL> SELECT ENAME FROM EMP WHERE SUBSTR(ENAME,3,1) = 'A';  
  
ENAME  
-----  
BHANU  
  
SQL> SELECT ENAME FROM EMP WHERE ENAME LIKE '__A%';  
  
ENAME  
-----  
BHANU
```

Question 15

Display the last name of all employees who have both an A and an S in their ename.

Query:

```
SELECT ENAME FROM EMP WHERE ENAME LIKE '%A%' AND ENAME LIKE '%S%';
```

Output

```
SQL> SELECT ENAME FROM EMP WHERE ENAME LIKE '%A%' AND ENAME LIKE '%S%';

ENAME
-----
ASHU
JAMES
AISHU
SAM
```

Question 16

Display the Ename, Job, Sal for all employees whose jobs are CLERK and whose salary is in 800 or 950 or 1300.

Query:

```
SELECT ENAME,JOB,SAL FROM EMP WHERE JOB='CLERK'AND SAL IN (800,950,1300);
```

Output:

```
SQL> SELECT ENAME,JOB,SAL FROM EMP WHERE JOB='CLERK'AND SAL IN (800,950,1300);

ENAME      JOB      SAL
-----
DEEPU      CLERK      800
JAMES      CLERK      950

SQL>
```